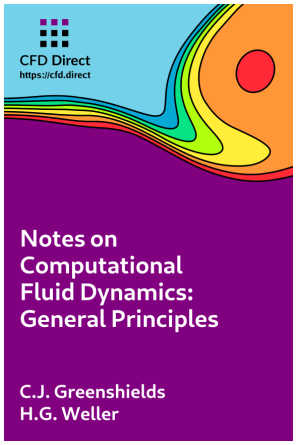
Notes on Computational Fluid Dynamics: General Principles



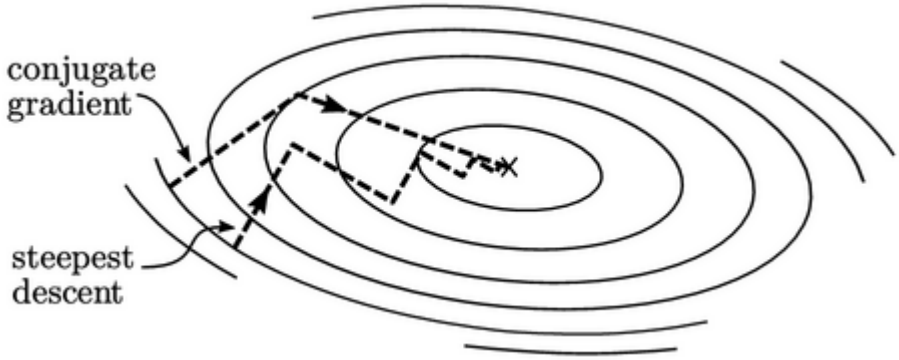
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5.15 Conjugate gradient method

The solution to a matrix equation $\mathbf{A} \bullet \Psi = \mathbf{b}$ by descent methods involves finding the minimum of the equation in quadratic form. This can be illustrated using the contours of the paraboloid describing the quadratic for the case with values Ψ_1 and Ψ_2.



The search for the minimum involves a series of updates to Ψ of the form

$$\Psi := \Psi^c + \alpha \mathbf{p} \tag{5.32}$$

where Ψ is the next update using latest (current) values Ψ^c. The column vector $\mathbf{p}$ provides the *direction* of the line search towards the minimum; the scalar α provides the magnitude of the line is that direction.

Steepest descent

The intuitive way to reach the minimum is to follow the direction of *steepest descent*. The method is fairly simple because the direction is defined by the negative of gradient of the quadratic form $\partial f(\Psi)/\partial \Psi$ which is the residual $\mathbf{r} = \mathbf{b} - \mathbf{A} \bullet \Psi$.

The distance to "walk" is naturally until the lowest point is reached, corresponding to[5]

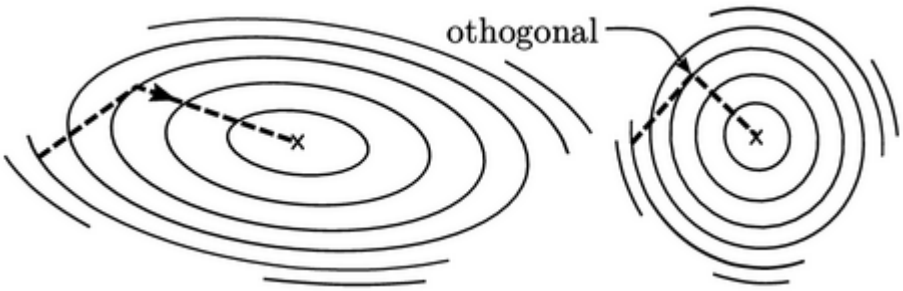
$$\alpha = \frac{\mathbf{r} \cdot \mathbf{r}}{\mathbf{r} \cdot \mathbf{A} \cdot \mathbf{r}}. \tag{5.33}$$

The method reaches the minimum point along a zigzag path where consecutive directions are orthogonal. The directions do not change and result in a *larger* number of very short steps, the *more* the paraboloid is stretched in one direction.

The conjugate direction

The **conjugate gradient** (CG) method chooses search directions that are *conjugate* with $\mathbf{A}$. This means each new direction $\mathbf{p}$ corresponds to the previous one $\mathbf{p}^c$, satisfying

$$\mathbf{p}^c \cdot \mathbf{A} \cdot \mathbf{p} = 0. \tag{5.34}$$



This can be imagined as the directions being orthogonal *with the stretch in the paraboloid removed*. For 2 values, CG finds the minimum in 2 steps, rather than several zigzag steps.

CG provides the basis for practical matrix solvers for CFD, described in Sec. 5.16. For a detailed explanation of the CG method, see the reference below.

[5] Jonathan Richard Shewchuk *An introduction to the conjugate gradient method without the agonizing pain*, 1994.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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